

1. What is the address of GPIOF_BSRR?
 - a. 0x48001814
 - b. 0x48001828
 - c. 0x48001418
 - d. 0x48001028

2. To light up an active-low LED interfaced to Port B Pin 12, you must write value to address.....
 - a. 0x10000000, 0x48000418
 - b. 0x00001000, 0x48000428
 - c. 0x10000000, 0x48000428
 - d. 0x00001000, 0x48000418

3. To configure Port F Pin 14 as I2C4_SCL, you must write.....to..... register.
 - a. 4'b1000 to GPIOF AFRL
 - b. 4'b0100 to GPIOF AFRH
 - c. 4'b1000 to GPIOF AFRH
 - d. 4'b0100 to GPIOF AFRL

4. Assume STM32 in its new chip design introduces a new register GPIOx_BTR.

GPIOx_BTR (Address offset = 0x30): Bit Toggle Register

GPIOx_BTR BTy, for port x pin y (y = 15:0) works as below.

These bits are write-only, a read to these bits returns the value 0x0000.

Bits 31:16 of this register are reserved.

- 0: no action on the corresponding ODy bit
- 1: Toggles the corresponding ODy bit

BT31	BT30	BT29	BT28	BT27	BT26	BT25	BT24	BT23	BT22	BT21	BT20	BT19	BT18	BT17	BT16
BT15	BT14	BT13	BT12	BT11	BT10	BT9	BT8	BT7	BT6	BT5	BT4	BT3	BT2	BT1	BT0
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

- a) Write the new HAL_GPIO_TogglePin function by using the new BTR register.
- b) What is the address of GPIOD_BTR?
- c) Two LEDs (PB6 and PB10) are to be toggled together every second. What would be the value written to the GPIOB_BTR register every second?